



SO Margin Guard

User Manual for Odoo 19.0

Module version: 19.0.1.1.0

Purpose: enforce margin policy, require authorized reasons, and preserve an immutable audit trail.

Area	Behavior
Rule security	System Administrator or Margin Guard Rule Administrator only
Single confirmation	Allow, block, or request a mandatory override reason
Batch confirmation	Process compliant orders and request overrides only for affected orders
Audit log	Automatically created, read-only, and permanent
Rule changes	Effective immediately after save

1. Security and Access

- **System Administrator:** can create, edit, archive, and maintain rules.
- **Margin Guard Rule Administrator:** dedicated technical group for rule maintenance without granting general Sales Manager control.
- **Sales User / Sales Manager:** may read relevant information and logs, but cannot change rules unless separately authorized.
- **Override Group Member:** may override only the rules that point to a group containing that user.

Assign technical permission from **Settings > Users & Companies > Groups**, then add the user to **Margin Guard Rule Administrator**.

2. Creating a Rule

Open **Margin Guard > Rules**. Complete the target model, guarded method, threshold, optional domain, formula, override group, reason settings, and error message.

Field	Recommended value for Sales Orders
Model	Sales Order (sale.order)
Method to Guard	action_confirm
Domain	[] to include all orders
Minimum Margin %	Company policy threshold, for example 10.0
Override Group	Authorized sales management group
Require Reason	Enabled for audit-controlled overrides

Example error message

This %(model)s is below the required margin %(actual).1f%% < %(threshold).1f%%. Override blocked.

Allowed placeholders are **%(model)s**, **%(actual).1f**, and **%(threshold).1f**. Use **%%** for a literal percent sign. Invalid templates are rejected while saving the rule.

3. Immediate Rule Activation

Active rules become effective immediately after saving. The current Odoo worker installs a dynamic wrapper for a new model and method, while active rule values are read from the database on every guarded call. Other workers are signaled to refresh automatically.

A rule cannot be saved as active when its model or method does not exist. Domain format, formula syntax, thresholds, reason length, and error-message placeholders are validated before save.

4. Single-Order Confirmation

- Margin meets the threshold: confirmation runs normally.
- Margin is below the threshold and the user lacks override permission: confirmation is blocked with the configured message.
- Margin is below the threshold and the user is authorized: the reason wizard opens when a reason is required.
- The audit log must be created successfully before the final confirmation can complete.

5. Batch Confirmation

When multiple records are selected, every record is evaluated. The module separates them into three groups:

- **Compliant:** processed normally immediately.
- **Authorized override required:** included in one batch reason wizard.
- **Blocked:** left unconfirmed and listed clearly because the current user cannot override all violated rules.

Only records that actually need a reason are sent to the wizard. The wizard also identifies records already processed normally and any blocked records. Cancelling the wizard leaves affected records unconfirmed while compliant records remain processed.

6. Immutable Margin Override Log

Open **Margin Guard > Override Log**. Each entry stores the approving user, time, rule, source record, threshold, calculated margin, gap, and reason.

- Logs cannot be created manually.
- Logs cannot be edited, duplicated, or deleted through normal access.
- The model enforces immutability in Python in addition to read-only access rights and views.
- A rule with existing audit logs cannot be deleted. Archive it instead.
- If audit creation fails, the override confirmation is cancelled.

7. Functional Test Checklist

1. Create an active sale.order / action_confirm rule with a threshold of 50.
2. Choose an Override Group and require a reason of at least 10 characters.
3. Create one order above the threshold and one below it.
4. Select both orders and click Confirm.
5. Verify that the compliant order confirms and only the low-margin order appears in the wizard.
6. Enter a valid reason and confirm the wizard.
7. Verify the second order confirms and exactly one audit log is created for the violation.
8. Try to edit or delete the log and verify that both operations are rejected.

8. Troubleshooting

Rule cannot be saved: correct the model, method, domain, formula syntax, threshold, reason length, or error-message placeholders.

Wizard does not appear: confirm that the user belongs to every Override Group required by the violated rules.

Some batch records remain quotations: read the wizard or warning message; it lists records without override permission.

Audit logging failure: review the Odoo server log and database permissions. Confirmation is intentionally stopped to protect compliance.